

1. Effect of different types of activation function in 10 CNNs

Accuracy Comparison Table

Model	Activation	Transfer Accuracy	Fine-tune Accuracy
MobileNet	Softmax	0.7355	0.7415
	ReLU	0.6635	0.2175
	tanh	0.0935	0.0500
VGG16	Softmax	0.6040	0.7955
	ReLU	0.4455	0.1280
	tanh	0.0610	0.0785
ResNet50	Softmax	0.0850	0.1290
	ReLU	0.1115	0.1165
	tanh	0.0470	0.0580
InceptionV3	Softmax	0.6075	0.6540
	ReLU	0.5415	0.4825
	tanh	0.1095	0.0500
InceptionResNetV2	Softmax	0.6685	0.6755
	ReLU	0.5325	0.5400
	tanh	0.0955	0.0390
MobileNetV2	Softmax	0.7195	0.7360
	ReLU	0.6175	0.6490
	tanh	0.1280	0.0905
DenseNet121	Softmax	0.7270	0.7290
	ReLU	0.6330	0.6480
	tanh	0.1030	0.0975

In our 20-class classification task using the CIFAR-100 dataset, we experimented with different activation functions in the output layer: SoftMax, ReLU, and Tanh. We observed that using **SoftMax** resulted in the highest accuracy. This is expected, as SoftMax is specifically designed for multi-class classification—it transforms raw output scores into probabilities that sum to one, making it ideal for selecting a single class from multiple options. When we applied **ReLU** in the output layer, the model's accuracy dropped significantly. ReLU does not normalize outputs or provide probabilistic interpretation, and it may zero out negative values, which can impair learning. **Tanh** performed even worse, likely due to its output range of -1 to 1, which can cause vanishing gradients and hinder training stability. We retained **ReLU** in the hidden layers due to

its efficiency and ability to maintain strong gradient flow. Based on these results, we concluded that **SoftMax** is the most suitable activation function for the output layer in multi-class classification tasks.

2. Kernel Type Usage in 10 CNNs

Kernel Type	Models That Use It
Regular Kernel (3×3, 5×5)	VGG16, ResNet50, DenseNet121
Deformable Kernel	Not in standard versions; extended variants of ResNet and Inception
Dilated Kernel	InceptionV3, InceptionResNetV2 (also in DeepLab variants)
Depthwise Separable Kernel	MobileNet, MobileNetV2, Xception
Modified Depthwise-Separable	MobileNetV2 (inverted residuals), NASNetMobile
Pointwise Kernel (1×1)	MobileNet, Xception, InceptionV3, ResNet50

3. Feature Map Analysis Across Layers

Model	Early Layers	Mid Layers	Late Layers
Mobile Net	Basic edges, corners	Object parts like wheels, eyes	Full object patterns (e.g., face, car)
VGG16	Horizontal/vertical edges	Shapes from edge combinations	Complex textures, full object representations
ResNet50	Shallow edge detectors	Feature fusion via residuals, mid-level shapes	Deep semantic-rich object representations
Inception v3	Multi-scale filter extraction	Mixed resolutions, context capture	Aggregated fine-grained multi-scale features
InceptionResNet V2	Diverse low-level features via mixed paths	Identity + convolution paths	Abstract representations from residual-inception fusion
Mobile NetV2	Edge maps and basic textures	Compressed (bottlenecked) mid-level features	Efficient, class-specific embeddings
DenseNet121	Simple filters shared across layers	Collective learned features through dense connectivity	Very dense, semantic feature representations
DenseNet201	Similar to DenseNet121	Deeper network improves feature reuse granularity	Finer, more discriminative representations

Xception	Directional features via depthwise convolutions	Channel-wise independent feature maps	Fused spatial and channel-domain features
NASNet Mobile	Low-level patterns via Swish and separable convs	Neural-architecture-optimized combinations	Class-specific abstraction with efficient computation